Test Plan and Test Cases

Warehouse Inventory Application - exam answer format

1. AIM

To design a test plan and test cases for a warehouse inventory application covering stock receipt, stock issue, inventory adjustment, supplier details and stock reconciliation, and to execute them.

2. REQUIREMENTS

Type	Details
Hardware	Computer with minimum 4 GB RAM
Software	Python 3.x, IDLE or VS Code, Windows operating system

3. TEST PLAN

Heading	Content
Objective	Verify that the 5 modules work correctly and reject invalid input
Scope	Supplier details, stock receipt, stock issue, inventory adjustment, stock reconciliation
Approach	Black-box testing, positive and negative testing, boundary value analysis
Test data	Items I1 = 100 and I2 = 20; supplier S1
Entry criteria	Code is ready and test cases are designed
Exit criteria	All test cases executed and no test case failed
Risks	Negative stock, duplicate receipt, wrong variance

4. PROCEDURE

1. Create the item stock, the supplier list and a set of used receipt references.
2. Write one function for each module: add_supplier, receive, issue, adjust and reconcile.
3. Write a check() function that compares the actual result with the expected result and prints PASS or FAIL.
4. Call check() once for every test case.
5. Print the total, passed and failed counts.
6. Run the program and note the output.

5. TEST CASES

Initial data: I1 = 100, I2 = 20, supplier S1.

ID	Module	Input	Expected result
TC01	Supplier	Add S2, name XYZ	OK
TC02	Supplier	Add S1 again (duplicate)	REJECTED
TC03	Supplier	Blank supplier id	REJECTED
TC04	Receipt	I1, S1, quantity 25, reference G1	OK
TC05	Receipt	Stock of I1 after the receipt	125
TC06	Receipt	Same reference G1 again	REJECTED
TC07	Receipt	Quantity 0	REJECTED
TC08	Receipt	Unknown supplier S9	REJECTED
TC09	Issue	I1, quantity 30	OK
TC10	Issue	Stock of I1 after the issue	95
TC11	Issue	I2, quantity 21 (stock is 20)	REJECTED
TC12	Issue	I2, quantity 20 (stock is 20)	OK
TC13	Adjustment	I1, +5, reason 'count'	OK
TC14	Adjustment	I1, -5, no reason	REJECTED
TC15	Adjustment	I2, -1 (stock is 0)	REJECTED
TC16	Reconciliation	I1, physical count 98 (system stock 100)	Variance -2
TC17	Reconciliation	Physical count -1	REJECTED

6. PROGRAM

```
# Warehouse Inventory - Test Cases (exam version)

stock = {"I1": 100, "I2": 20}        # item -> quantity
suppliers = {"S1": "ABC Traders"}    # supplier id -> name
grn_used = set()                     # receipt references already used

def add_supplier(sid, name):
    if not sid or not name or sid in suppliers:
        return "REJECTED"
    suppliers[sid] = name
    return "OK"

def receive(item, sid, qty, grn):
    if item not in stock or sid not in suppliers or qty <= 0 or grn in grn_used:
        return "REJECTED"
    stock[item] += qty
    grn_used.add(grn)
    return "OK"

def issue(item, qty):
    if item not in stock or qty <= 0 or qty > stock[item]:
        return "REJECTED"
    stock[item] -= qty
    return "OK"

def adjust(item, change, reason):
    if item not in stock or change == 0 or not reason or stock[item] + change < 0:
        return "REJECTED"
    stock[item] += change
    return "OK"

def reconcile(item, physical):
    if item not in stock or physical < 0:
        return "REJECTED"
    variance = physical - stock[item]
    stock[item] = physical
    return variance

passed = 0
total = 0

def check(tid, desc, actual, expected):
    global passed, total
    total += 1
    result = "PASS" if actual == expected else "FAIL"
    if result == "PASS":
        passed += 1
    print(tid, desc, "| Expected:", expected, "| Actual:", actual, "|", result)

# ---------------- Test cases ----------------
check("TC01", "Add valid supplier",        add_supplier("S2", "XYZ"), "OK")
check("TC02", "Duplicate supplier id",     add_supplier("S1", "Dup"), "REJECTED")
check("TC03", "Blank supplier id",         add_supplier("", "XYZ"), "REJECTED")
check("TC04", "Valid receipt of 25",       receive("I1", "S1", 25, "G1"), "OK")
check("TC05", "Stock after receipt",       stock["I1"], 125)
check("TC06", "Duplicate receipt ref",     receive("I1", "S1", 10, "G1"), "REJECTED")
check("TC07", "Receipt of zero quantity",  receive("I1", "S1", 0, "G2"), "REJECTED")
check("TC08", "Receipt, unknown supplier", receive("I1", "S9", 5, "G3"), "REJECTED")
check("TC09", "Valid issue of 30",         issue("I1", 30), "OK")
check("TC10", "Stock after issue",         stock["I1"], 95)
check("TC11", "Issue 21 from stock 20",    issue("I2", 21), "REJECTED")
check("TC12", "Issue exactly 20 of 20",    issue("I2", 20), "OK")
check("TC13", "Adjust +5 with reason",     adjust("I1", 5, "count"), "OK")
check("TC14", "Adjust without reason",     adjust("I1", -5, ""), "REJECTED")
check("TC15", "Adjust below zero",         adjust("I2", -1, "loss"), "REJECTED")
check("TC16", "Reconcile, physical = 98",  reconcile("I1", 98), -2)
check("TC17", "Reconcile, negative count", reconcile("I1", -1), "REJECTED")

print("Total:", total, "Passed:", passed, "Failed:", total - passed)
```

7. OUTPUT

```
TC01 Add valid supplier | Expected: OK | Actual: OK | PASS
TC02 Duplicate supplier id | Expected: REJECTED | Actual: REJECTED | PASS
TC03 Blank supplier id | Expected: REJECTED | Actual: REJECTED | PASS
TC04 Valid receipt of 25 | Expected: OK | Actual: OK | PASS
TC05 Stock after receipt | Expected: 125 | Actual: 125 | PASS
TC06 Duplicate receipt ref | Expected: REJECTED | Actual: REJECTED | PASS
TC07 Receipt of zero quantity | Expected: REJECTED | Actual: REJECTED | PASS
TC08 Receipt, unknown supplier | Expected: REJECTED | Actual: REJECTED | PASS
TC09 Valid issue of 30 | Expected: OK | Actual: OK | PASS
TC10 Stock after issue | Expected: 95 | Actual: 95 | PASS
TC11 Issue 21 from stock 20 | Expected: REJECTED | Actual: REJECTED | PASS
TC12 Issue exactly 20 of 20 | Expected: OK | Actual: OK | PASS
TC13 Adjust +5 with reason | Expected: OK | Actual: OK | PASS
TC14 Adjust without reason | Expected: REJECTED | Actual: REJECTED | PASS
TC15 Adjust below zero | Expected: REJECTED | Actual: REJECTED | PASS
TC16 Reconcile, physical = 98 | Expected: -2 | Actual: -2 | PASS
TC17 Reconcile, negative count | Expected: REJECTED | Actual: REJECTED | PASS
Total: 17 Passed: 17 Failed: 0
```

8. RESULT

A test plan and 17 test cases were designed for the warehouse inventory application and executed successfully. All 17 test cases passed, so the stock receipt, stock issue, inventory adjustment, supplier details and stock reconciliation modules work as expected.

Exam tips

1. If time is short, write only 10 rows of the test case table (1 or 2 for each module).
2. Copy the program exactly, with the same indentation. Run it and write the output you get.
3. Viva: TC11 and TC12 are boundary value analysis, TC02 and TC06 are negative testing, TC04 is positive testing, TC16 is stock reconciliation (variance = physical - system).